

2(a)(i)	<i>Mycobacterium tuberculosis</i> or <i>Mycobacterium bovis</i> ;		1
2(a)(ii)	artificial active immunity	✓ ;	1
	artificial passive immunity		
	natural active immunity		
	natural passive immunity		
2(b)(i)	<i>three from</i> changes the tertiary structure / shape of active site, of , enzyme / RNA polymerase ; <u>active site</u> no longer complementary to, RNA nucleotides / substrate ; AW RNA nucleotides not joined / mRNA not synthesised ; A transcription does not occur translation prevented ; A described, I polypeptide / protein synthesis prevented AVP ; <i>ref. to</i> other roles of RNA polymerase not occurring, e.g. does not bind to DNA, double helix does not unwind		3
2b(ii)	<i>two from</i> (from 2009 to 2013) overall increase in number of cases (of RR-TB and MDR-TB) ; use of data to support overall trend ; <i>ref. to</i> large increase between 2011 and 2012 / 2013 or very little change between 2010 and 2011 ;		2

Question	Answer	Marks
2(b)(iii)	<i>four from</i> overuse of, antibiotics / rifampicin or example of over use ; e.g. taking for viral infection, over prescribing for bacterial infection people not completing course of antibiotics ; reservoir of bacteria remains ; <i>ref. to</i> mutation ; any detail of the mutation ; e.g. protein produced has a changed binding site bacteria with resistance, survive / selected for or only bacteria sensitive to antibiotic are, killed / selected against ; A antibiotic acts as a selection pressure bacteria reproduce and pass on, gene / allele, for resistance to offspring or vertical (gene) transmission ; (alleles for resistance to antibiotics transferred by) horizontal (gene) transmission / described ; frequency of resistance, gene / allele, increases in the bacterial population ;	4